

May 2024

ME1003-2 – ELEMENTS OF MECHANICAL ENGINEERING
(CV)

Max. Marks: 100

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

20 Marks

1. **The slope of the stress-strain curve in the elastic deformation region is**
A) Elastic modulus
B) Plastic modulus
C) Poisson's ratio
D) Bulk modulus
2. **In hydroelectric power plant**
A) Kinetic energy is transferred to potential
B) Potential energy of water is converted into Electrical energy
C) Solar energy is transferred to wind energy
D) Wind energy is transferred to solar energy
3. **3D printing means**
A) Creating two-dimensional designs
B) Adding layers to create three-dimensional objects
C) Printing text documents on paper
D) Printing holographic images
4. **Trapped heat inside the earth is known as**
A) Heat energy
B) Kinetic energy
C) Geothermal energy
D) Thermal energy
5. **Centrifugal pumps are used to transport**
A) Pressure
B) Speed
C) Power
D) Liquids
6. **A centrifugal compressor works on the principle of**
A) Conversion of pressure energy into kinetic energy
B) Conversion of kinetic energy into pressure energy
C) Centripetal action
D) Generating pressure directly
7. **In an impulse steam turbine**
A) The steam is expanded in nozzles only and there is a pressure drop and heat drop
B) The steam is expanded both in fixed and moving blades continuously
C) The steam is expanded in moving blades only
D) The pressure and temperature of steam remains constant
8. **A pipe conveying water from Dam to water Turbine**
A) Penstock
B) Tailstock
C) Runner
D) Stator
9. **The ratio of brake power to indicated power is called**
A) Indicated thermal efficiency
B) Relative efficiency
C) Mechanical efficiency
D) Volumetric efficiency
10. **Which of the following medium is compressed in a diesel engine cylinder?**
A) Air alone
B) Air and fuel
C) Air and lubrication oil
D) Fuel alone
11. **In a vapor compression cycle, where do we find the lowest temperature?**
A) Evaporator
B) Condenser
C) Expansion Valve
D) Compressor

12. What does a Vapor Compression refrigerator use in the form of a refrigerant?
 - A) Aqua ammonia
 - B) Freon
 - C) Ammonia
 - D) Water
13. Worm gear is used, when two shafts are at
 - A) Right angles to each other
 - B) Some distance apart
 - C) Radial direction
 - D) 45 cm apart
14. Helical gear is used for transmitting the power between two _____
 - A) Parallel shafts
 - B) Inclined shafts
 - C) Shafts located at long distance
 - D) Shafts at 90°
15. Electronic components are joined by which of the following methods?
 - A) Spot Welding
 - B) Seam Welding
 - C) Soldering
 - D) Brazing
16. Which joint formed by the following processes is the strongest.
 - A) Brazing
 - B) Soldering
 - C) Gluing
 - D) Welding
17. Which of the following operation does NOT use a rotating tool?
 - A) Reaming
 - B) Counter-Boring
 - C) Facing
 - D) End milling
18. On drilling machine, which process is known as reaming?
 - A) Enlargement of existing hole
 - B) Hole made by removal of metal along the hole circumference
 - C) Smoothly finishing and accurately sizing a drilled hole
 - D) Reducing the diameter of drilled hole by adding material inside.
19. Structures such as gear, linkages, and belts fall under which of the following types of components in a mechatronics system?
 - A) Electrical component
 - B) Electronic component
 - C) Computer Control system
 - D) Mechanical components
20. Which one of the following engineering deals with controlling and intelligence of robots?
 - A) Electrical
 - B) Mechanical
 - C) Computer
 - D) Architecture

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the role of mechanical engineering in industry and society.	4	L*1	1	1
	b) Write a brief note on additive manufacturing.	4	L1	1	1
	c) Define the meaning of a prime mover. Explain the working of Kaplan reaction water turbine with a suitable figure.	8	L2	2	1
2.	a) Explain the process formation of steam with suitable sketches and highlighting various terminologies involved.	8	L1	1	1
	b) Illustrate the working of Impulse Steam turbine (De-Laval) with the help of necessary sketches.	8	L2	2	1
3.	a) Define the meaning of a power plant. Explain the working of a nuclear power plant with the help of a sketch.	8	L2	1	1
	b) Define the meaning of a compressor. Explain the working of a single stage air compressor with help of a sketch.	8	L2	2	1
Unit – II					
4.	a) Describe the working of a 4-stroke diesel engine with necessary figures showing all strokes and P-V diagram.	8	L2	3	1
	b) A shaft running at 100 rpm is to drive a parallel shaft at 150 rpm. The pulley on the driving shaft is 35cm in diameter. Find the diameter of the driven pulley. Calculate the linear velocity of the belt and also the velocity ratio.	6	L3	4	1
	c) Differentiate between simple and compound gear trains.	2	L1	4	1

5. a) The following observations were made during a trial on a 4-stroke diesel engine.
 Cylinder diameter = 25 cm; Stroke of the piston = 40 cm;
 Crankshaft speed = 250 rpm; Brake load = 70 kg;
 Brake drum diameter = 2 m; Mean effective pressure = 6 bar;
 Disel oil consumption = 0.1 m³/min; Specific gravity of diesel = 0.78;
 Calorific value of petrol = 43,900 kJ/kg.
 Find: (i) Indicated power (ii) Brake power
 (iii) Mechanical efficiency (iv) Brake thermal efficiency. 8 L3 3 1
 b) Define Gear. Explain different types of gears with suitable sketches. 8 L1 4 1
6. a) Explain the working of electric and hybrid vehicles with the help of block diagrams. 8 L2 3 1
 b) Define welding. Explain the method of TIG welding with a suitable sketch. 8 L2 4 1

Unit – III

7. a) Explain the following.
 i) Fixed automation
 ii) Programmable automation
 iii) Flexible automation
 iv) Open and closed loop control systems. 8 L1 5 1
 b) Explain the following Lathe operations.
 i) Facing ii) Turning
 iii Thread Cutting. iv) Knurling 8 L1 5 1
8. a) Define Robotics. Also list the applications of robots. 4 L1 5 1
 b) Briefly explain with sketches the classification robots based on configuration. 4 L1 5 1
 c) Explain the following machining operations with neat sketches.
 i) Drilling
 ii) Tapping
 iii) Plane milling
 iv) Slot milling. 8 L1 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
